**Supplementary Material**

**for: DWTA-FL: Drift-Weighted Trust-Aware Federated Learning for Robust Multi-Cloud Intrusion Detection**

All tables in this supplement were regenerated for the revision. Tables S7–S9 are unchanged and are the single source for every detection number in the main text.

**Table S1.** Retained input features of the final training matrix for each dataset. The label, attack-category, IP-address, timestamp and flow-ID columns are removed.

**CSE-CIC-IDS2018** — 78 retained input features (removed: Timestamp, Label). The Flow ID, Src IP, Src Port and Dst IP columns that appear in one daily file are also removed. All 78 features are numerical.

| | | |
|---|---|---|
| 1. Dst Port | 2. Protocol | 3. Flow Duration |
| 4. Tot Fwd Pkts | 5. Tot Bwd Pkts | 6. TotLen Fwd Pkts |
| 7. TotLen Bwd Pkts | 8. Fwd Pkt Len Max | 9. Fwd Pkt Len Min |
| 10. Fwd Pkt Len Mean | 11. Fwd Pkt Len Std | 12. Bwd Pkt Len Max |
| 13. Bwd Pkt Len Min | 14. Bwd Pkt Len Mean | 15. Bwd Pkt Len Std |
| 16. Flow Byts/s | 17. Flow Pkts/s | 18. Flow IAT Mean |
| 19. Flow IAT Std | 20. Flow IAT Max | 21. Flow IAT Min |
| 22. Fwd IAT Tot | 23. Fwd IAT Mean | 24. Fwd IAT Std |
| 25. Fwd IAT Max | 26. Fwd IAT Min | 27. Bwd IAT Tot |
| 28. Bwd IAT Mean | 29. Bwd IAT Std | 30. Bwd IAT Max |
| 31. Bwd IAT Min | 32. Fwd PSH Flags | 33. Bwd PSH Flags |
| 34. Fwd URG Flags | 35. Bwd URG Flags | 36. Fwd Header Len |
| 37. Bwd Header Len | 38. Fwd Pkts/s | 39. Bwd Pkts/s |
| 40. Pkt Len Min | 41. Pkt Len Max | 42. Pkt Len Mean |
| 43. Pkt Len Std | 44. Pkt Len Var | 45. FIN Flag Cnt |
| 46. SYN Flag Cnt | 47. RST Flag Cnt | 48. PSH Flag Cnt |
| 49. ACK Flag Cnt | 50. URG Flag Cnt | 51. CWE Flag Count |
| 52. ECE Flag Cnt | 53. Down/Up Ratio | 54. Pkt Size Avg |
| 55. Fwd Seg Size Avg | 56. Bwd Seg Size Avg | 57. Fwd Byts/b Avg |
| 58. Fwd Pkts/b Avg | 59. Fwd Blk Rate Avg | 60. Bwd Byts/b Avg |
| 61. Bwd Pkts/b Avg | 62. Bwd Blk Rate Avg | 63. Subflow Fwd Pkts |
| 64. Subflow Fwd Byts | 65. Subflow Bwd Pkts | 66. Subflow Bwd Byts |
| 67. Init Fwd Win Byts | 68. Init Bwd Win Byts | 69. Fwd Act Data Pkts |

| | | |
|---|---|---|
| 70. Fwd Seg Size Min | 71. Active Mean | 72. Active Std |
| 73. Active Max | 74. Active Min | 75. Idle Mean |
| 76. Idle Std | 77. Idle Max | 78. Idle Min |

**UNSW-NB15** — 42 retained input features (removed: srcip, sport, dstip, Stime, Ltime, attack_cat, Label). proto, state and service are categorical and are one-hot encoded after the split; dsport is kept as a numerical service indicator.

| | | |
|---|---|---|
| 1. dsport | 2. proto | 3. state |
| 4. dur | 5. sbytes | 6. dbytes |
| 7. sttl | 8. dttl | 9. sloss |
| 10. dloss | 11. service | 12. Sload |
| 13. Dload | 14. Spkts | 15. Dpkts |
| 16. swin | 17. dwin | 18. stcpb |
| 19. dtcpb | 20. smeansz | 21. dmeansz |
| 22. trans_depth | 23. res_bdy_len | 24. Sjit |
| 25. Djit | 26. Sintpkt | 27. Dintpkt |
| 28. tcprtt | 29. synack | 30. ackdat |
| 31. is_sm_ips_ports | 32. ct_state_ttl | 33. ct_flw_http_mthd |
| 34. is_ftp_login | 35. ct_ftp_cmd | 36. ct_srv_src |
| 37. ct_srv_dst | 38. ct_dst_ltm | 39. ct_src_ltm |
| 40. ct_src_dport_ltm | 41. ct_dst_sport_ltm | 42. ct_dst_src_ltm |

**TON_IoT** — 40 retained input features (removed: ts, src_ip, dst_ip, label, type). The previously reported count of 44 included a target column. The categorical columns (proto, service, conn_state, dns_query, ssl_*, http_* strings, weird_*) are one-hot or frequency encoded after the split.

| | | |
|---|---|---|
| 1. src_port | 2. dst_port | 3. proto |
| 4. service | 5. duration | 6. src_bytes |
| 7. dst_bytes | 8. conn_state | 9. missed_bytes |
| 10. src_pkts | 11. src_ip_bytes | 12. dst_pkts |
| 13. dst_ip_bytes | 14. dns_query | 15. dns_qclass |
| 16. dns_qtype | 17. dns_rcode | 18. dns_AA |
| 19. dns_RD | 20. dns_RA | 21. dns_rejected |
| 22. ssl_version | 23. ssl_cipher | 24. ssl_resumed |

| | | |
|---|---|---|
| 25. ssl_established | 26. ssl_subject | 27. ssl_issuer |
| 28. http_trans_depth | 29. http_method | 30. http_uri |
| 31. http_version | 32. http_request_body_len | 33. http_response_body_len |
| 34. http_status_code | 35. http_user_agent | 36. http_orig_mime_types |
| 37. http_resp_mime_types | 38. weird_name | 39. weird_addl |
| 40. weird_notice | | |

**Table S2.** Every original label and its binary mapping

| **Dataset** | **Original label (label column)** | **Binary class** | **Code** |
|---|---|---|---|
| CSE-CIC-IDS2018 (Label) | Benign | Benign | 0 |
| | Bot | Malicious | 1 |
| | Brute Force -Web | Malicious | 1 |
| | Brute Force -XSS | Malicious | 1 |
| | SQL Injection | Malicious | 1 |
| | DDOS attack-HOIC | Malicious | 1 |
| | DDOS attack-LOIC-UDP | Malicious | 1 |
| | DDoS attacks-LOIC-HTTP | Malicious | 1 |
| | DoS attacks-GoldenEye | Malicious | 1 |
| | DoS attacks-Hulk | Malicious | 1 |
| | DoS attacks-SlowHTTPTest | Malicious | 1 |
| | DoS attacks-Slowloris | Malicious | 1 |
| | FTP-BruteForce | Malicious | 1 |
| | SSH-Bruteforce | Malicious | 1 |
| | Infilteration (spelling as in the dataset) | Malicious | 1 |
| UNSW-NB15 (attack_cat; Label) | Normal (empty attack_cat, Label = 0) | Benign | 0 |
| | Fuzzers | Malicious | 1 |
| | Analysis | Malicious | 1 |
| | Backdoor / Backdoors | Malicious | 1 |
| | DoS | Malicious | 1 |
| | Exploits | Malicious | 1 |
| | Generic | Malicious | 1 |
| | Reconnaissance | Malicious | 1 |
| | Shellcode | Malicious | 1 |

| Dataset | Original label (label column) | Binary class | Code |
|---|---|---|---|
| | Worms | Malicious | 1 |
| TON_IoT (type; label) | normal | Benign | 0 |
| | backdoor | Malicious | 1 |
| | ddos | Malicious | 1 |
| | dos | Malicious | 1 |
| | injection | Malicious | 1 |
| | mitm | Malicious | 1 |
| | password | Malicious | 1 |
| | ransomware | Malicious | 1 |
| | scanning | Malicious | 1 |
| | xss | Malicious | 1 |

*Note.* Leading and trailing spaces in the raw UNSW-NB15 attack_cat strings are stripped before mapping. Repeated header rows inside the CSE-CIC-IDS2018 files are removed during cleaning.

**Table S3.** Benign and malicious training samples of every client (Dirichlet $\alpha = 0.5$, partition seed 42; training partition only)

| Client | CSE-CIC benign | CSE-CIC malicious | UNSW benign | UNSW malicious | TON_IoT benign | TON_IoT malicious |
|---|---|---|---|---|---|---|
| 1 | 190,975 | 205 | 91,553 | 31,906 | 10,058 | 3,492 |
| 2 | 897,173 | 11,613 | 217,550 | 1 | 12,233 | 64 |
| 3 | 33,138 | 16,850 | 2,008 | 5,501 | 2,594 | 516 |
| 4 | 4,592 | 52,278 | 105,973 | 453 | 46,673 | 11,789 |
| 5 | 511,510 | 625,953 | 111,346 | 79,303 | 1,484 | 6,568 |
| 6 | 576 | 419,872 | 38,458 | 8,509 | 22,448 | 2,502 |
| 7 | 1,491,332 | 64,311 | 90 | 4,174 | 6,278 | 280 |
| 8 | 45,007 | 1,358 | 138 | 7,806 | 193 | 3,318 |
| 9 | 126,013 | 354 | 13,879 | 12,279 | 38,093 | 17,978 |
| 10 | 254,002 | 26,200 | 253,729 | 83 | 2,563 | 14,103 |
| 11 | 534,535 | 186,031 | 23,655 | 57,349 | 14,300 | 1 |
| 12 | 116,191 | 13,687 | 7,352 | 662 | 34,519 | 1,347 |
| 13 | 283,167 | 3,444 | 11,791 | 15,959 | 582 | 24,087 |
| 14 | 54,277 | 963 | 308,024 | 1,782 | 229 | 637 |

| Client | CSE-CIC benign | CSE-CIC malicious | UNSW benign | UNSW malicious | TON_IoT benign | TON_IoT malicious |
|---|---|---|---|---|---|---|
| 15 | 525,161 | 138,350 | 60,290 | 4,269 | 0 | 8,988 |
| 16 | 5,760 | 5 | 131,448 | 13,558 | 6,778 | 211 |
| 17 | 4,602,778 | 102,018 | 226,121 | 257 | 10,349 | 17,185 |
| 18 | 126,322 | 137,690 | 134,095 | 3,248 | 13,437 | 2,923 |
| 19 | 725,085 | 228,435 | 14,203 | 52 | 2,361 | 144 |
| 20 | 20,275 | 16,771 | 7,297 | 8,283 | 10,248 | 10,632 |
| **Total** | **10,547,869** | **2,046,388** | **1,759,000** | **255,434** | **235,420** | **126,765** |

*Note.* Training-set totals (80 % of the retained records in main-text Table 4): CSE-CIC-IDS2018 12,594,257 (benign 10,547,869 + malicious 2,046,388); UNSW-NB15 2,014,434 (benign 1,759,000 + malicious 255,434); TON_IoT 362,185 (benign 235,420 + malicious 126,765). Each column pair sums exactly to its dataset total, and each class total equals the class count of the training split. All per-client cells are exported directly from the saved seed-42 Dirichlet ($\alpha = 0.5$) partition file (client index lists) with the script released with the code; they are not estimated or rescaled.

**Table S4.** Performance under increasing synthetic drift on CSE-CIC-IDS2018 (mean of five runs; drift schedule in Table 5 of the main text)

| Drift level | Method | Accuracy (%) | Precision (%) | Recall (%) | F1 (%) | FPR (%) | Global loss |
|---|---|---|---|---|---|---|---|
| 0% | FedAvg | 96.93 | 97.61 | 97.48 | 97.54 | 1.21 | 0.086 |
| 0% | FLTrust | 98.06 | 97.89 | 97.82 | 97.85 | 1.08 | 0.079 |
| 0% | FedSBS | 98.17 | 98.03 | 97.96 | 97.99 | 0.99 | 0.075 |
| 0% | FedMADE | 98.24 | 98.11 | 98.04 | 98.07 | 0.94 | 0.071 |
| 0% | DWTA-FL | 98.42 | 98.31 | 98.25 | 98.28 | 0.82 | 0.064 |
| 10% | FedAvg | 96.91 | 96.64 | 96.43 | 96.53 | 1.74 | 0.104 |
| 10% | FLTrust | 97.48 | 97.21 | 97.05 | 97.13 | 1.43 | 0.092 |
| 10% | FedSBS | 97.71 | 97.48 | 97.29 | 97.38 | 1.29 | 0.087 |
| 10% | FedMADE | 97.86 | 97.62 | 97.47 | 97.54 | 1.18 | 0.081 |
| 10% | DWTA-FL | 98.16 | 97.94 | 97.83 | 97.88 | 0.97 | 0.071 |
| 20% | FedAvg | 95.48 | 95.07 | 94.82 | 94.94 | 2.51 | 0.137 |
| 20% | FLTrust | 96.43 | 96.12 | 95.86 | 95.98 | 2.02 | 0.116 |
| 20% | FedSBS | 96.82 | 96.51 | 96.28 | 96.39 | 1.82 | 0.107 |
| 20% | FedMADE | 97.04 | 95.88 | 96.57 | 96.22 | 1.69 | 0.101 |
| 20% | DWTA-FL | 97.63 | 97.35 | 97.18 | 97.26 | 1.38 | 0.084 |

| Drift level | Method | Accuracy (%) | Precision (%) | Recall (%) | F1 (%) | FPR (%) | Global loss |
|---|---|---|---|---|---|---|---|
| 30% | FedAvg | 93.87 | 93.34 | 92.96 | 93.15 | 3.42 | 0.181 |
| 30% | FLTrust | 95.21 | 94.77 | 94.42 | 94.59 | 2.79 | 0.151 |
| 30% | FedSBS | 95.73 | 95.31 | 94.96 | 95.13 | 2.51 | 0.139 |
| 30% | FedMADE | 96.08 | 95.67 | 95.34 | 95.50 | 2.34 | 0.129 |
| 30% | DWTA-FL | 96.91 | 96.54 | 96.27 | 96.40 | 1.91 | 0.104 |

*Note. Every F1 equals 2PR/(P+R) of the precision and recall in its row. FedMADE at 20% drift is F1 96.22, the harmonic mean of precision 95.88 and recall 96.57. The 0% rows come from the windowed drift protocol in main-text Table 5. They are not the same runs as main-text Table 9, so the 0% DWTA-FL row is not required to equal F1 99.06% or FPR 1.16%.*

**Table S5.** Adversarial attack configuration (malicious clients cannot read or modify server-side statistics)

| Attack | Malicious clients | Configuration |
|---|---|---|
| Label poisoning | 20% | 20% of local samples; label flipped (benign ↔ malicious) |
| Sign flipping | 20% | Update direction reversed; scale = 3 |
| Backdoor | 20% | 10% of local rows. CSE-CIC-IDS2018: Protocol=0, Dst Port=0, Flow Duration=0. UNSW-NB15: proto=unas, service=-, state=INT. TON_IoT: proto=tcp, service=-, conn_state=OTH. Label set to the target class benign (0). |
| Byzantine | 20% | Random update with 3× the legitimate update L2 norm |
| Adaptive Byzantine | 20% | Adversarial direction chosen each round; norm ≤ 3× benign median |
| Collaborative collusion | 10%, 20%, 30% | Shared adversarial direction; scale = 3 |
| Covert gradient escape | 10%, 20%, 30% | Targeted update; ≤ 15% deviation from the benign update norm |
| Extreme stress | 50% | Coordinated manipulation |
| Attack period | – | Rounds 20–100 |

**Table S6a.** Mean ± SD over the five seeds of Tables S7–S9 (SD with the $n - 1$ denominator)

| Dataset | Method | Accuracy (%) | F1 (%) | FPR (%) |
|---|---|---|---|---|
| CSE-CIC-IDS2018 | FLEX-IDS | 96.84 ± 0.53 | 96.42 ± 0.57 | 3.06 ± 0.18 |
| | FL-IDS++ | 97.63 ± 0.42 | 97.14 ± 0.46 | 2.41 ± 0.24 |
| | FL-IDPP | 96.97 ± 0.56 | 96.45 ± 0.64 | 2.86 ± 0.16 |

| Dataset | Method | Accuracy (%) | F1 (%) | FPR (%) |
|---|---|---|---|---|
| | Fed-ANIDS | 95.74 ± 0.78 | 95.18 ± 0.72 | 3.63 ± 0.28 |
| | DWTA-FL | 98.42 ± 0.26 | 99.06 ± 0.24 | 1.16 ± 0.13 |
| UNSW-NB15 | FLEX-IDS | 96.52 ± 0.61 | 96.13 ± 0.55 | 3.36 ± 0.23 |
| | FL-IDS++ | 97.21 ± 0.52 | 96.81 ± 0.46 | 2.56 ± 0.22 |
| | FL-IDPP | 96.74 ± 0.66 | 96.37 ± 0.59 | 3.08 ± 0.29 |
| | Fed-ANIDS | 95.63 ± 0.68 | 95.05 ± 0.59 | 3.74 ± 0.24 |
| | DWTA-FL | 98.37 ± 0.33 | 98.75 ± 0.24 | 1.50 ± 0.16 |
| TON_IoT | FLEX-IDS | 96.31 ± 0.69 | 95.94 ± 0.52 | 3.39 ± 0.28 |
| | FL-IDS++ | 97.08 ± 0.39 | 96.72 ± 0.36 | 2.50 ± 0.19 |
| | FL-IDPP | 96.63 ± 0.63 | 96.27 ± 0.59 | 3.08 ± 0.27 |
| | Fed-ANIDS | 95.48 ± 0.82 | 94.96 ± 0.61 | 3.77 ± 0.29 |
| | DWTA-FL | 98.11 ± 0.35 | 98.46 ± 0.26 | 1.55 ± 0.15 |

**Table S6b.** Paired tests of DWTA-FL against each baseline (five paired seeds)

| Dataset | Baseline | Δ Accuracy, points [95% CI] | Δ F1, points [95% CI] | Exact Wilcoxon p | Holm p | $r_{rb}$ |
|---|---|---|---|---|---|---|
| CSE-CIC-IDS2018 | FLEX-IDS | 1.58 [1.19, 1.97] | 2.64 [2.21, 3.07] | 0.0625 | 0.25 | 1.00 |
| | FL-IDS++ | 0.79 [0.52, 1.06] | 1.92 [1.62, 2.22] | 0.0625 | 0.25 | 1.00 |
| | FL-IDPP | 1.45 [1.05, 1.85] | 2.61 [2.10, 3.12] | 0.0625 | 0.25 | 1.00 |
| | Fed-ANIDS | 2.68 [1.95, 3.41] | 3.88 [3.19, 4.57] | 0.0625 | 0.25 | 1.00 |
| UNSW-NB15 | FLEX-IDS | 1.85 [1.43, 2.27] | 2.62 [2.15, 3.08] | 0.0625 | 0.25 | 1.00 |
| | FL-IDS++ | 1.16 [0.91, 1.41] | 1.94 [1.61, 2.27] | 0.0625 | 0.25 | 1.00 |
| | FL-IDPP | 1.63 [1.19, 2.07] | 2.38 [1.90, 2.86] | 0.0625 | 0.25 | 1.00 |
| | Fed-ANIDS | 2.74 [2.29, 3.19] | 3.70 [3.26, 4.14] | 0.0625 | 0.25 | 1.00 |
| TON_IoT | FLEX-IDS | 1.80 [1.37, 2.23] | 2.52 [2.19, 2.85] | 0.0625 | 0.25 | 1.00 |
| | FL-IDS++ | 1.03 [0.74, 1.32] | 1.74 [1.51, 1.97] | 0.0625 | 0.25 | 1.00 |
| | FL-IDPP | 1.48 [1.07, 1.89] | 2.19 [1.73, 2.65] | 0.0625 | 0.25 | 1.00 |
| | Fed-ANIDS | 2.63 [2.01, 3.25] | 3.50 [2.92, 4.08] | 0.0625 | 0.25 | 1.00 |

*Note.* Δ = DWTA-FL minus baseline; each 95% CI is a t interval (4 degrees of freedom) for the mean paired difference of the metric named in its column. Exact two-sided Wilcoxon signed-rank test on five pairs: when all five differences have the same sign, $p = 2/2^5 = 0.0625$, the smallest value possible with $n = 5$. Holm adjustment within each dataset and metric over four baselines (ordered p-values multiplied by 4, 3, 2 and 1, then running maximum); with four tied raw values of 0.0625 every adjusted value is 0.25. The p-values are the same for accuracy and F1. Rank-biserial correlation $r_{\mathrm{rb}} = (W^+ - W^-)/(W^+ + W^-)$.

**Table S7. Individual Random-Seed Results — CSE-CIC-IDS2018**

| Method | Seed | Accuracy (%) | F1-Score (%) | FPR (%) |
|---|---|---|---|---|
| FLEX-IDS | 101 | 96.12 | 95.71 | 3.28 |
| FLEX-IDS | 202 | 97.31 | 96.91 | 2.91 |
| FLEX-IDS | 303 | 96.74 | 96.38 | 3.04 |
| FLEX-IDS | 404 | 97.42 | 97.06 | 2.87 |
| FLEX-IDS | 505 | 96.61 | 96.04 | 3.19 |
| FL-IDS++ | 101 | 97.14 | 96.63 | 2.62 |
| FL-IDS++ | 202 | 98.02 | 97.51 | 2.19 |
| FL-IDS++ | 303 | 97.38 | 96.91 | 2.48 |
| FL-IDS++ | 404 | 98.11 | 97.73 | 2.13 |
| FL-IDS++ | 505 | 97.50 | 96.92 | 2.64 |
| FL-IDPP | 101 | 96.31 | 95.92 | 3.05 |
| FL-IDPP | 202 | 97.46 | 96.92 | 2.73 |
| FL-IDPP | 303 | 96.82 | 96.48 | 2.91 |
| FL-IDPP | 404 | 97.63 | 97.21 | 2.67 |
| FL-IDPP | 505 | 96.63 | 95.72 | 2.94 |
| Fed-ANIDS | 101 | 94.82 | 94.21 | 3.94 |
| Fed-ANIDS | 202 | 96.31 | 95.74 | 3.42 |

| Method | Seed | Accuracy (%) | F1-Score (%) | FPR (%) |
|---|---|---|---|---|
| Fed-ANIDS | 303 | 95.21 | 94.68 | 3.81 |
| Fed-ANIDS | 404 | 96.73 | 95.92 | 3.26 |
| Fed-ANIDS | 505 | 95.63 | 95.35 | 3.72 |
| **DWTA-FL** | 101 | 98.21 | 98.86 | 1.29 |
| **DWTA-FL** | 202 | 98.71 | 99.31 | 1.04 |
| **DWTA-FL** | 303 | 98.36 | 98.97 | 1.18 |
| **DWTA-FL** | 404 | 98.68 | 99.32 | 1.02 |
| **DWTA-FL** | 505 | 98.14 | 98.84 | 1.27 |

**Table S8. Individual Random-Seed Results — UNSW-NB15**

| Method | Seed | Accuracy (%) | F1-Score (%) | FPR (%) |
|---|---|---|---|---|
| FLEX-IDS | 101 | 95.74 | 95.42 | 3.62 |
| FLEX-IDS | 202 | 96.83 | 96.31 | 3.28 |
| FLEX-IDS | 303 | 96.41 | 96.03 | 3.41 |
| FLEX-IDS | 404 | 97.36 | 96.92 | 3.02 |
| FLEX-IDS | 505 | 96.26 | 95.99 | 3.47 |
| FL-IDS++ | 101 | 96.58 | 96.31 | 2.81 |
| FL-IDS++ | 202 | 97.84 | 97.42 | 2.37 |
| FL-IDS++ | 303 | 97.06 | 96.68 | 2.59 |
| FL-IDS++ | 404 | 97.63 | 97.15 | 2.31 |
| FL-IDS++ | 505 | 96.94 | 96.50 | 2.72 |
| FL-IDPP | 101 | 95.93 | 95.61 | 3.46 |

| Method | Seed | Accuracy (%) | F1-Score (%) | FPR (%) |
|---|---|---|---|---|
| FL-IDPP | 202 | 97.38 | 97.02 | 2.83 |
| FL-IDPP | 303 | 96.72 | 96.34 | 3.11 |
| FL-IDPP | 404 | 97.41 | 96.87 | 2.76 |
| FL-IDPP | 505 | 96.26 | 96.01 | 3.22 |
| Fed-ANIDS | 101 | 94.71 | 94.13 | 4.12 |
| Fed-ANIDS | 202 | 96.28 | 95.61 | 3.51 |
| Fed-ANIDS | 303 | 95.44 | 94.89 | 3.83 |
| Fed-ANIDS | 404 | 96.32 | 95.47 | 3.57 |
| Fed-ANIDS | 505 | 95.40 | 95.15 | 3.68 |
| **DWTA-FL** | 101 | 98.01 | 98.42 | 1.61 |
| **DWTA-FL** | 202 | 98.74 | 99.02 | 1.37 |
| **DWTA-FL** | 303 | 98.22 | 98.61 | 1.52 |
| **DWTA-FL** | 404 | 98.69 | 98.93 | 1.31 |
| **DWTA-FL** | 505 | 98.19 | 98.77 | 1.69 |

**Table S9.** Individual Random-Seed Results — TON_IoT

| Method | Seed | Accuracy (%) | F1-Score (%) | FPR (%) |
|---|---|---|---|---|
| FLEX-IDS | 101 | 95.61 | 95.28 | 3.71 |
| FLEX-IDS | 202 | 96.83 | 96.34 | 3.19 |
| FLEX-IDS | 303 | 96.24 | 95.82 | 3.46 |
| FLEX-IDS | 404 | 97.18 | 96.57 | 3.02 |
| FLEX-IDS | 505 | 95.69 | 95.69 | 3.58 |

| Method | Seed | Accuracy (%) | F1-Score (%) | FPR (%) |
|---|---|---|---|---|
| FL-IDS++ | 101 | 96.64 | 96.31 | 2.76 |
| FL-IDS++ | 202 | 97.62 | 97.13 | 2.28 |
| FL-IDS++ | 303 | 96.83 | 96.42 | 2.54 |
| FL-IDS++ | 404 | 97.31 | 97.01 | 2.36 |
| FL-IDS++ | 505 | 97.00 | 96.73 | 2.58 |
| FL-IDPP | 101 | 95.82 | 95.41 | 3.39 |
| FL-IDPP | 202 | 97.31 | 96.92 | 2.81 |
| FL-IDPP | 303 | 96.47 | 96.08 | 3.16 |
| FL-IDPP | 404 | 97.22 | 96.73 | 2.79 |
| FL-IDPP | 505 | 96.33 | 96.21 | 3.25 |
| Fed-ANIDS | 101 | 94.52 | 93.98 | 4.21 |
| Fed-ANIDS | 202 | 96.27 | 95.43 | 3.49 |
| Fed-ANIDS | 303 | 95.18 | 94.76 | 3.87 |
| Fed-ANIDS | 404 | 96.41 | 95.32 | 3.54 |
| Fed-ANIDS | 505 | 95.02 | 95.31 | 3.74 |
| **DWTA-FL** | 101 | 97.81 | 98.18 | 1.72 |
| **DWTA-FL** | 202 | 98.43 | 98.71 | 1.43 |
| **DWTA-FL** | 303 | 98.06 | 98.39 | 1.57 |
| **DWTA-FL** | 404 | 98.51 | 98.76 | 1.36 |
| **DWTA-FL** | 505 | 97.74 | 98.26 | 1.65 |

*Note to Tables S7–S9.* Seeds 101, 202, 303, 404 and 505 were used for every method; runs with the same seed share the client partition (seed 42) and form one paired observation.

**Table S10. Per-dataset attack results. Main-text Tables 14–17 report CSE-CIC-IDS2018; the same attacks on UNSW-NB15 and TON_IoT are reported here.**

*CSE-CIC-IDS2018* — accuracy (%) / ARR (%) under collusion (10%, 20%, 30% malicious), adaptive Byzantine (20%) and extreme stress (50%)

| Method | Collusion 10% | Collusion 20% | Collusion 30% | Adaptive Byzantine | Extreme stress |
|---|---|---|---|---|---|
| FedAvg | 81.2 / 74.5 | 71.4 / 62.1 | 58.6 / 49.3 | 65.9 | 63.8 / 58.4 |
| FLTrust | 91.4 / 88.2 | 84.6 / 81.0 | 76.8 / 71.4 | 76.8 | 79.6 / 75.1 |
| Krum | 92.1 / 89.1 | 86.2 / 82.5 | 78.9 / 73.2 | 78.1 | 81.3 / 77.8 |
| Bulyan | 93.0 / 90.4 | 87.5 / 83.9 | 80.1 / 75.6 | 79.3 | 82.4 / 79.1 |
| DWTA-FL | 97.8 / 96.2 | 94.6 / 92.8 | 90.8 / 88.4 | 86.8 | 89.2 / 86.7 |

*UNSW-NB15* — accuracy (%) / ARR (%) under collusion (10%, 20%, 30% malicious), adaptive Byzantine (20%) and extreme stress (50%)

| Method | Collusion 10% | Collusion 20% | Collusion 30% | Adaptive Byzantine | Extreme stress |
|---|---|---|---|---|---|
| FedAvg | 79.8 / 72.1 | 69.5 / 59.8 | 56.4 / 47.1 | 63.2 | 61.5 / 55.8 |
| FLTrust | 90.1 / 86.9 | 83.2 / 79.4 | 74.9 / 69.2 | 75.1 | 77.8 / 73.0 |
| Krum | 91.0 / 87.8 | 84.9 / 81.0 | 77.1 / 71.3 | 76.4 | 79.2 / 75.2 |
| Bulyan | 91.8 / 88.9 | 86.1 / 82.2 | 78.5 / 73.4 | 77.9 | 80.5 / 76.9 |
| DWTA-FL | 97.2 / 95.4 | 93.8 / 91.6 | 89.4 / 86.8 | 85.2 | 88.1 / 85.0 |

*TON_IoT* — accuracy (%) / ARR (%) under collusion (10%, 20%, 30% malicious), adaptive Byzantine (20%) and extreme stress (50%)

| Method | Collusion 10% | Collusion 20% | Collusion 30% | Adaptive Byzantine | Extreme stress |
|---|---|---|---|---|---|
| FedAvg | 78.5 / 70.4 | 68.1 / 58.2 | 55.0 / 45.8 | 61.8 | 60.1 / 54.2 |
| FLTrust | 89.4 / 85.6 | 81.9 / 78.1 | 73.5 / 67.8 | 73.9 | 76.4 / 71.5 |
| Krum | 90.2 / 86.5 | 83.5 / 79.7 | 75.8 / 69.9 | 75.0 | 78.0 / 73.8 |
| Bulyan | 91.1 / 87.6 | 84.8 / 80.9 | 77.2 / 71.8 | 76.5 | 79.2 / 75.4 |
| DWTA-FL | 96.8 / 94.7 | 93.1 / 90.8 | 88.5 / 85.6 | 84.4 | 87.2 / 83.9 |

**Table S11a.** Ablation on UNSW-NB15 (same protocol as main-text Table 23)

| Variant | Accuracy (%) | F1 (%) | ARR (%) | FPR (%) |
|---|---|---|---|---|
| Full DWTA-FL | 98.37 | 98.75 | 94.2 | 1.50 |

| Variant | Accuracy (%) | F1 (%) | ARR (%) | FPR (%) |
|---|---|---|---|---|
| − Drift penalty | 96.81 | 96.65 | 87.8 | 3.12 |
| − Recursive trust | 96.12 | 95.94 | 84.1 | 3.65 |
| − Consistency | 95.70 | 95.31 | 82.2 | 4.08 |
| − Drift and trust | 94.45 | 94.10 | 77.6 | 5.05 |
| − Drift and consistency | 93.94 | 93.52 | 75.9 | 5.48 |
| − Trust and consistency | 93.05 | 92.64 | 72.1 | 6.42 |

*Note. The full-model accuracy, F1 and FPR are the five-seed means of Table S6. The other rows are the ablation variants for this dataset. They are not pooled with another dataset.*

**Table S11b.** Ablation on TON_IoT (same protocol as main-text Table 23)

| Variant | Accuracy (%) | F1 (%) | ARR (%) | FPR (%) |
|---|---|---|---|---|
| Full DWTA-FL | 98.11 | 98.46 | 93.6 | 1.55 |
| − Drift penalty | 96.54 | 96.30 | 87.1 | 3.30 |
| − Recursive trust | 95.82 | 95.60 | 83.4 | 3.84 |
| − Consistency | 95.41 | 95.02 | 81.5 | 4.29 |
| − Drift and trust | 94.12 | 93.75 | 76.8 | 5.30 |
| − Drift and consistency | 93.60 | 93.18 | 75.0 | 5.75 |
| − Trust and consistency | 92.71 | 92.25 | 71.4 | 6.70 |

*Note. The full-model accuracy, F1 and FPR are the five-seed means of Table S6. The other rows are the ablation variants for this dataset. They are not pooled with another dataset.*

**Section S-B. Details of the bounded-stationarity sketch (main text, Section 3.10)**

*Aggregate noise with cross-client covariance.* With $\xi^t = \bar{g}^t - \mathbb{E}[\bar{g}^t \mid \mathcal{F}_t]$ and $\bar{g}^t = \sum_i \alpha_i^t g_i^t$, if the weights are treated as fixed given $\mathcal{F}_t$,

$$\mathbb{E}[\| \xi^t \|_2^2 \mid \mathcal{F}_t] = \sum_i (\alpha_i^t)^2 \sigma_i^2 + \sum_{i \neq j} \alpha_i^t \alpha_j^t \operatorname{tr} \operatorname{Cov}(g_i^t, g_j^t) \leq \sigma_{\text{agg}}^2,$$

so no independence between clients is assumed. Because the weights depend on the updates, this expression is an approximation; Assumption (A3) bounds the true conditional second moment directly.

*Inner-product bound (Step 2).* Since $\mathbb{E}[\xi^t \mid \mathcal{F}_t] = 0$, $\mathbb{E}[\langle \nabla F, \bar{g}^t \rangle \mid \mathcal{F}_t] = \| \nabla F \|_2^2 + \langle \nabla F, r^t \rangle$. Young's inequality gives $\langle \nabla F, r^t \rangle \geq -1/2 \| \nabla F \|_2^2 - 1/2 \| r^t \|_2^2 \geq -1/2 \| \nabla F \|_2^2 - 1/2 B_t^2$.

*Second-moment bound (Step 2).* $\mathbb{E}[\| \bar{g}^t \|_2^2 \mid \mathcal{F}_t] = \| \nabla F + r^t \|_2^2 + \mathbb{E}[\| \xi^t \|_2^2 \mid \mathcal{F}_t] \leq 2 \| \nabla F \|_2^2 + 2B_t^2 + \sigma_{\text{agg}}^2$.

*Step 3.* Substituting both bounds into the descent inequality gives $\mathbb{E}[F(w^{t+1}) \mid \mathcal{F}_t] \leq F(w^t) - (\eta/2 - L\eta^2) \| \nabla F \|_2^2 + (\eta/2 + L\eta^2) B_t^2 + L\eta^2/2\, \sigma_{\text{agg}}^2$; with $\eta \leq 1/(4L)$, $L\eta^2 \leq \eta/4$, which yields main-text Step 3.

*Step 4.* Taking total expectation, summing over $t = 0, \ldots, T-1$, using $F(w^T) \geq F_*$ and dividing by $\eta T/4$ gives main-text Step 4.

*Scope.* Drift and volatility are bounded by construction ($0 \leq \widehat{D}_i^t \leq 1$, $0 \leq V_i^t \leq 2$); these bounds only guarantee that the weights are well defined. $B_H$, $B_W$, $G_A$, $\bar{\rho}^t$ and $\sigma_{\text{agg}}^2$ are assumed, not estimated. The sketch models one gradient step per round and does not cover five local Adam epochs.